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Classical Cryptography

Substitution ciphers

Replace plaintext characters
according to a translation table.



Cryptanalysis



- For large key spaces brute forcing is no longer practical
 - Note: it is *unfeasible*, not impossible
- **Cryptanalysis** is the process of trying to work out the encryption key from a given ciphertext.
 - This type of attack exploits the characteristics of the algorithm (cipher) to attempt to deduce a specific plaintext or key
 - Attacker will analyze the ciphertext using different statistical tests – they must have prior knowledge of the nature of plaintext (plain English, photo, exe file etc.)

Cryptanalysis Techniques



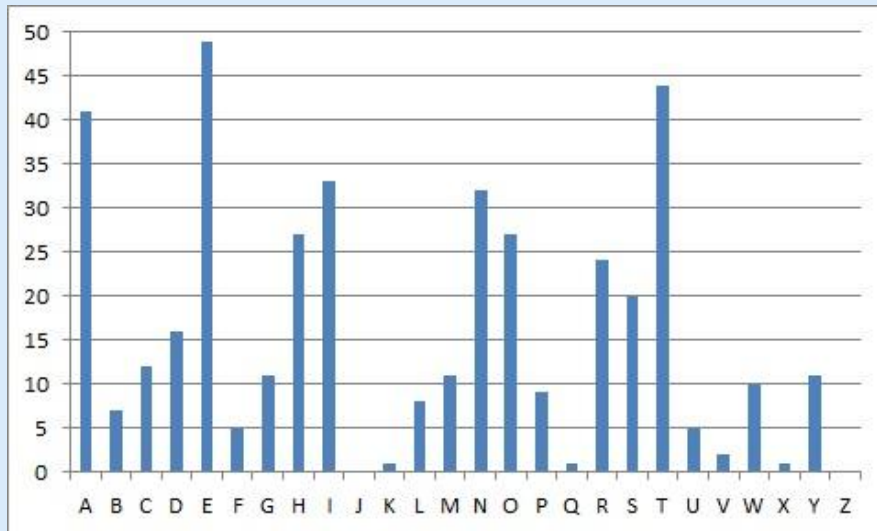
Frequency Analysis

- Compute frequencies of letters in ciphertext and compare with the typical frequencies in the target language.
 - e.g. in English texts, most frequent letters are E (13%), T (9.1%), A (8.2%), O (7.5%)
- Can also find frequency of groups of letters (di-grams and tri-grams): *an, in, the*

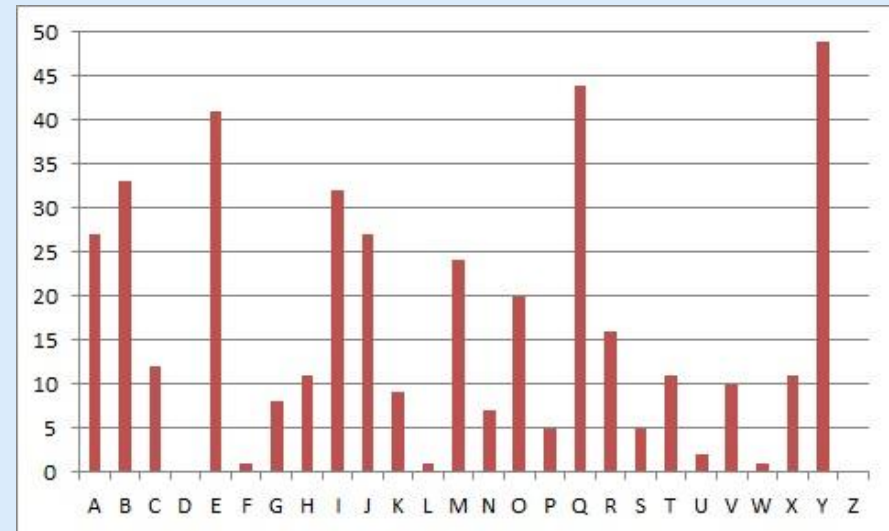
Cryptanalysis Techniques



Frequency Analysis Example 1



Plaintext
(English)



Ciphertext

Cryptanalysis Techniques



Frequency Analysis Example 2

UKBYBIPOUZBCUFEEBORUKBYBHOBBERFESPVKBWFOFERNBCVBZPRUBOFERNBCVBPCYY
FVUFOFEIKNWFRFIKJNUPWRFIPOUNVNIPUBRNCUKBEFWWFDNCHXCYBOHOPYXPUBNCU
BOYNRVNIWNCPOJIOFHOPZRVFZIXUBORJRUBZRBCHNCBBONCHRJZSFWNVJRUBZRPCYZ
PUKBZPUNVPWPCYVFZIXUPUNFCPWVRVNBCVBRPYYNUNFCPWWJUKBYBIPOUZBCUIPOUN
VNIPUBRNCHOPYXPUBNCUBOYNRVNIWNCPOJIOFHOPZRNCRVNBCUNENVVFZIXUNCHPCY
VFZIXUPUNFCPWZPUKBZPUNVR

B	36	→ E
N	34	→ T
U	33	→ A
P	32	
C	26	

NC	11	→ IN
PU	10	→ AT
UB	10	
UN	9	

Di-grams

UK	6	→ THE
B		
RV	6	
N		
FZI	4	

Tri-grams

Hardening the Substitution Cipher



- Monoalphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet.
- One possible countermeasure is to provide multiple substitutes, known as **homophones** , for a single letter.
 - e.g. letter e could be randomly assigned a number of different cipher symbols, such as 16, 74, 35, and 21
- If the number of symbols assigned to each letter is proportional to the relative frequency of that letter, then single-letter frequency information is completely obliterated.
- What about digram & trigram frequencies?

Hardening the Substitution Cipher



- To create a stronger cipher, we look into **polyalphabetic** substitution
- Most well-known example: **Vigenère Cipher**
- Uses Caesar's cipher with various different shifts, in order to hide the distribution of the letters.
- A 'keyword' defines the shifts used for each letter in the plaintext

Vigenère Cipher



- A keyword defines the shifts used in each letter in the text
- The key word is repeated many times to match the length of plaintext

Plaintext = “I attack you”

Keyword = “BOX”

(which translates to Ceaser shifts of “1, 14, 23”)

plaintext	I	a	t	t	a	c	k	y	o	u
key	1	14	23	1	14	23	1	14	23	1
ciphertext	J	o	q	u	o	z	l	m	l	v

Vigenère Cipher



Vigenère square is helpful when working by hand.

Header row is for plaintext.

First column represents the key letter.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Cryptanalysis of Vigenère cipher



- Strength of Vigenère cipher is that there are multiple ciphertext letters for each plaintext letter, one for each unique letter of the keyword.
 - e.g. In the preceding example, plaintext 't' can be mapped to 'u', 'h' or 'o' (with the shifts of 1, 14 and 23 respectively).
- Thus, the letter frequency information is obscured.
- However, not all knowledge of the plaintext structure is lost.
- If attacker finds out the key length, say n , they can split the ciphertext in to n sections, and apply frequency analysis on each section separately.

Cryptanalysis of Vigenère cipher



- For example, if key length = 3 is known, attacker can split the ciphertext in to 3 sections, and apply frequency analysis on each section separately.

ciphertext	J	o	q	u	o	z	l	m	l	v
Section 1	J			u			l			v
Section 2		o			o			m		
Section 3			q			z			l	

Cryptanalysis of Vigenère cipher



Guessing the key length: Kasiski's method

- Find repeated strings in the ciphertext. Their distance is expected to be a multiple of the key length.
- Example:

P. Text: TOBENOTORTOBE
Keyword: 1231231231231
C. Text: UQEFPRUQUUQEF

Diagraph	First Position	Second Position	Distance	Factors

Cryptanalysis of Vigenère cipher



Guessing the key length: Kasiski's method

- Find repeated strings in the ciphertext. Their distance is expected to be a multiple of the key length.
- Example:

P. Text: TOBENOTORTOBE

Keyword: 1231231231231

C. Text: UQEFPRUQUUQEF

Diagraph	First Position	Second Position	Distance	Factors
UQ	1	7	6	2, 3, 6
UQ	7	10	3	3

Cryptanalysis of Vigenère cipher



Guessing the key length: Kasiski's method

- Find repeated strings in the ciphertext. Their distance is expected to be a multiple of the key length.
- Example:

P. Text: TOBENOTORTOBE
Keyword: 1231231231231
C. Text: UQ**EF**PRUQUUQ**EF**

Diagraph	First Position	Second Position	Distance	Factors
UQ	1	7	6	2, 3, 6
UQ	7	10	3	3
EF	3	12	9	3, 3, 9

Cryptanalysis of Vigenère cipher



Guessing the key length: Kasiski's method

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- Example:

P. Text: TOBENOTORTOBE
Keyword: 1231231231231
C. Text: UQEFPRUQUUQEF

Diagraph	First Position	Second Position	Distance	Factors
UQ	1	7	6	2, 3, 6
UQ	7	10	3	3
EF	3	12	9	3, 3, 9
QE	2	11	9	3, 3, 9

Cryptanalysis of Vigenère cipher



Guessing the key length: Kasiski's method

- Find repeated strings in the ciphertext. Their distance is expected to be a multiple of the key length.
- Example:

P. Text: TOBENOTORTOBE
Keyword: 1231231231231
C. Text: UQEFPRUQUUQEF

Guess of Key length
 $\gcd(6, 3, 9, 9) = 3$

Diagraph	First Position	Second Position	Distance	Factors
UQ	1	7	6	2, 3, 6
UQ	7	10	3	3
EF	3	12	9	3, 3, 9
QE	2	11	9	3, 3, 9

Hardening Vigenère Cipher



- As seen earlier, Vigenère cipher can be attacked by breaking ciphertext into n sections, and frequency analyzing each of those sections separately.
- But this approach won't work if the keyword length is equal or more than that of plaintext.
 - So the key need not be repeated
 - This the idea applied in **One-Time Pad** cipher

Permutation (transposition) ciphers

Changing order of characters in plaintext



Railfence cipher



- The plaintext is written down as a **sequence of diagonals and then read off as a sequence of rows.**
- Example: to encipher the message “**meet me after the party**” with a rail fence of depth 3, we write the following

```
      m           m           t           h           r
    e   t   e   f   e   t   e   a   t
      e           a           r           p           y
```

- The encrypted message is

mm**th****re****te****fe****te****ate****ar****py**

Railfence cipher: Strength



- The rail fence cipher is not very strong; the number of practical keys is small enough that a cryptanalyst can try them all by hand.

Columnar transposition ciphers



- In this cipher, message is written in a grid row by row, and read off column by column.
- In addition, order of columns is also permuted. This order becomes the key of the cipher

Message:
THIS IS TOP SECRET

Key:
3142

3	1	4	2
T	H	I	S
I	S	T	O
P	S	E	C
R	E	T	

or Key="GAME" which translates to 3142, as per the alphabetic ordering of these four letters: A, E, G, M

Ciphertext:
HSSE SOC TIPR ITET

key defines the order in which we read columns

Security of Permutation Ciphers



Question

- Are permutation ciphers susceptible to frequency analysis?

Security of Permutation Ciphers



Question

- Are permutation ciphers susceptible to frequency analysis?

Answer

- Ciphertext will have the same letter frequencies as the original plaintext. The attacker learns that a permutation cipher has been used.
- But apart from that, frequency analysis is of no help to attacker in breaking the cipher.

Security of Permutation Ciphers



- For the columnar transposition just shown, cryptanalysis is fairly straightforward and involves laying out the ciphertext in a matrix and playing around with column positions.
- The cipher can be made significantly more secure by performing more than one stage of transposition (i.e. re-encrypting the ciphertext).

References



- Stallings CNS, chap 3
- Vigenère cipher cracking:
<https://crypto.interactive-maths.com/kasiski-analysis-breaking-the-code.html>